

Trust

Human-Human

Human-Machine

Machine-Machine

Trust in AI

In a Social Media Context

Plus Human-Machine and Machine-Machine

Discussion for IP Crowd

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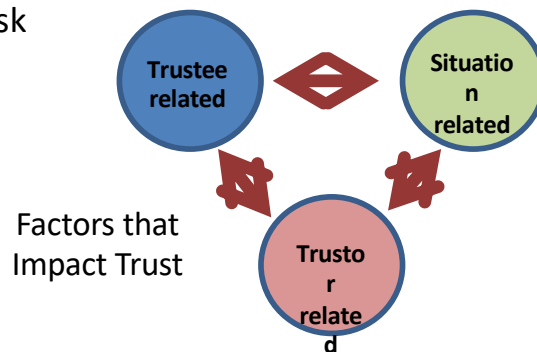
- Oxford Dictionary
 - trust **to believe that somebody is good, honest, sincere, etc. and that they will do what you expect of them or do the right thing**; to believe that something is true or correct: You can trust me not to tell anyone
- Trust exists between two entities—Trustor and Trustee
 - Trust: a belief, held by the trustor, that the trustee will act in a manner that mitigates the trustor's risk in a situation in which the trustee has put its outcomes at risk. [Wagner]
 - Overall, every definition of trust involves an individual in a position of vulnerability (the trustor) and a person on whom they must rely (the trustee)
 - the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor

Trust is a belief, held by the trustor (aggregator), that the trustee (sensor) will act in a manner that mitigates the trustor's risk in a situation in which the trustee has put its outcomes at risk

(Wagner, 2009)

Key Points

- Belief: mental statement about the world
- Held by trustor, impacting trustor, predicting trustee
- Risk



- Trust can be a single interaction or a durative relationship
- The trustor is the individual that must decide whether or not to put him or herself at risk or to be vulnerable to the actions of a trustee
- The trustee, may be a single individual, group of individuals, category of individuals, inanimate object, government institution, type of animal, etc

In our interconnected world

- Human-Human Trust
 - Trust of humans in person to person interaction
 - Trust of humans on social media
 - Online communications--- potential for misinformation
 - Everyone publishes online -- who to believe
 - E-commerce
- Human-Machine Trust (and Machine-Human Trust)
 - Human-Machine Teams
 - Human trusts of AI
 - Human trust of sensor readings and automated systems (e.g. GPS)
 - Military automated decisions
 - Automated pilot systems
 - Self-driving
 - Drones
 - E-commerce
- Machine-Machine Trust
 - For Example – Internet of things – Machines receiving input from external sensors
 - Energy grid, communications systems, infrastructure
 - E-commerce --- automated auctions --- for example bidding on power services, stock market

Factors or Dimensions in Trust

- Reputation
- Expertise – the importance of expertise --- trust an expert (vs perhaps a machine learning model)
- Consistency
- Past Experience– I have always had good service from a person or service provider
- Known uncertainties in measurements
 - Example – Internet of Things --- sensor can be embedded in the highway giving the temp to help drivers plan and avoid freezing conditions
 - GPS – known precision
 - Human-Human– example?
- Provenance – where do we think this entities information came from
- Perishability – how current is the information and how frequently does the information change
 - A good reputation 2 years ago may not still be valid
 - Technology upon which expertise is built might change over a short time
- Inheritance – DISCUSSION – how does inheritance play
 - A particular technology of environmental sensors is usually accurate
 - I am receiving feeds from temperature sensors embedded in the interstate in NC. Do they inherit the uncertainty levels of the sensors that I know about
 - Example – people who travel a lot often have good experience choosing restaurant
 - Sam travels a lot and may inherit this characteristic
 - When to apply inheritance properties

- Our definition of trust should be framed in the context of
 - the **set of decisions** that will be enabled by reasoning about the level of trust that the human has in a given human (H-M) (**decision-making module for M-M**) has in individual entities, sets of related entities or types of entities, providing information to the system or using or supplying services
 - How does this apply in H-H H-M M-M?
- The entity that I am trusting is very good at some things – not good at others
 - Ophthalmologist is an expert at Eye issues – I would not trust them with dental decisions
 - LL Bean has a good reputation for sporting clothing – would not necessarily trust them to provide high quality formal clothing
 - Whirlpool may have consistency in washing machine quality but their refrigerators may not have consistent quality
- Why does trust matter? What are you going to do with the information?
 - Decision?
 - Are you going to trust their information – or is there a high chance that it might be misinformation
 - Passing information (reputation)
 - Question for the class?

- How do you make decisions
- How should you design your app to make decisions
- Do you have a crowdsourcing app? How is it going to decide who to trust?
- How do you decide who to trust
- Yelp
- Reviews
- Reputation?
- Expertise?
- Discuss examples

- The other side of trust
- There is reason for an entity to trust you
- Behaves reliably
- Delivers good quality
- Having someone trust you does not do the person harm
- Based on expertise? Based on machine learning model trained on appropriate data? – is this trustworthy?
- Has a good reputation ? Is this trustworthy?

Building your trust in Social Media

How to know who to trust/ how to

- Cite reliable sources
- Don't be overly self-promotional
- Trust people or sites with appropriate expertise
- Reputation
- List provenance
- List perishability – point out and analyze
- If uncertainty is relevant – list the uncertainty in any measurements that your information is based on
- Consistency

- There is somewhat of a framework that exists around e-commerce
- Rating system – reputation- pre-existing data
- Human recommendations – How do you know which of the recommendations to trust?
- What are your comments on trust in e-commerce
- Yelp, consumer reports type comparisons

- We have listed a lot of dimensions of trust --I know you have been thinking about this --- what holds here?
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 - Inheritance

- B. Gebru *et al.*, "A Review on Human–Machine Trust Evaluation: Human-Centric and Machine-Centric Perspectives," in *IEEE Transactions on Human-Machine Systems*, vol. 52, no. 5, pp. 952-962, Oct. 2022, doi: 10.1109/THMS.2022.3144956.
- HMI requires the factors affecting trust to be properly accounted for and their relative importance to be rightly quantified.
- Trust is viewed from two perspectives—human-centric and machine-centric.
- The human trust - scales, and approaches, which are available to measure and calibrate human trust
- Machine trust --trustworthy artificial intelligence, built-in machine assurances, and ethical frameworks of trustworthy machines.
- Human-machine teaming – includes thinking about decision support apps – the role of trust

- Mis-information is a big problem on social media
- Political
- Covid
- Russian Trolls
- Email phishing
- There are techniques for recognizing trolls or phishing attempts – often based on wordings, misspellings, or stylometry -- style of writing
 - Stylometry has been used to analyze writings to identify the author

- Think of a reasoning system reasoning about the state of a system based on outputs from a network of sensors
- The features upon which the trust level of a node is based include
 - Uncertainty levels of measurements associated with the information coming from the node
 - Provenance: where does the information about this node come from?
 - Perishability: how current is the information that we have about this node and how quickly does the information change?
 - Past performance and reliability
- In addition, active learning techniques allow us to probe or query for pieces of information that would help to characterize the trust level of a node.
- Nodes will **inherit** trust features based on their node type or characteristics enabling a calculation of trust for new nodes as they become part of the network. A confidence level will be associated with the trust value assigned to a node and a level of risk will be assigned to any decision involving a node. The level of risk is based on the consequences or cost of misjudging the trust level of a node. Trust is dynamic and dependent on the decision which will be based upon that trust.
- **Inheritance** --- where does inheritance play in trust----
 - Human-human
 - In the past – shoe repair shops have given good advice on the best sandwich shops
 - Human –machine
 - Machine-machine
- DISCUSSION HERE

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- The level of risk is based on the consequences or cost of misjudging the trust level of a node.
- Trust is dynamic and dependent on the decision which will be based upon that trust.

- Explainable AI
 - Understanding the data
 - Understanding the algorithm
 - Understanding the context
 - Understanding the human expertise
 - Understanding –
-
- What is the risk of misjudging trust?

Why Expertise Matters-In deciding who to trust

- G. Klein, B. Shneiderman, R. R. Hoffman and K. M. Ford, "Why Expertise Matters: A Response to the Challenges," in *IEEE Intelligent Systems*, vol. 32, no. 6, pp. 67-73, November/December 2017, doi: 10.1109/MIS.2017.4531230.
- There are those who argue that expert knowledge is not necessary if there is big data which can be used to build a machine-learning based model that can be run to make predictions in a specific domain
- The above experts in psychology and human-systems interactions argue that Human Expertise is still important in deciding who to trust
- This paper argues that expertise matters in deciding who to trust
 - employ more effective strategies than others, and do so with less effort
 - perceive meaning in patterns that others do not notice
 - form rich mental models of situations to support sensemaking and anticipatory thinking
 - have extensive and highly organized domain knowledge
 - are intrinsically motivated to work on hard problems that stretch their capabilities.
- Society needs experts, even though they are fallible----
- Again – context matters --- what is the decision and domain – what is the history, reputation, consistency, quality, and past experience
- Provenance and perishability matter
 - E.g, how old is the human expertise – is it up to date
 - Where did the data that trained the machine learning algorithm come from

How can you talk to someone in a productive way

- Is this based on trust?
- Cognitive biases
 - This has to do with recognizing that there are innate psychological characteristics that are decision-making shortcuts that may make it difficult to change people's minds
 - Anchoring bias
 - Confirmation bias
 - Projection bias – don't assume that this person thinks like I do or likes what I like
 - Understanding the base rate – if in the population 85% prefer vanilla ice cream – the probability that this person prefers pistachio
- Not really getting to decisions
- Polite and productive conversations
- Productive (is this what you believe is productive)
 - Understand the person's beliefs
 - Understand what drives those beliefs
 - Understand that the situation is more complicated than we consider
 - Learn to communicate our beliefs
 - What drives our beliefs
 - What is the same and what is different between our beliefs and another person's
 - Trusting that the other person is trying to understand you
 - Trying to get you to change your beliefs in a non-malicious way
 - Understanding why they want you to understand their beliefs

- Trust is dynamic
- Reputations change
- The world of expertise changes – what is known
- The environment is changing – policies, behaviors, trends, consumer products, technologies
- Quality of service changes
- Information changes
- People automatically update their trust – but sometimes they need to be reminded to update
- Machine-Machine, Machine-Human need to be reminded to update
- Trust Algorithms need to be updated

Individual trust – as represented in your social media

- How can we build a platform that helps people trust in one another in appropriate ways
- Can we sculpt the interactions
- Templates interactions
- Workflows
- Reminders
- Monitors
- The act of writing comments– hints –prompts that would help the content be more trustworthy and less trollish
- Analogy to useability – to carl – prompt—there are cultures that might find that word offensive
- All of these different prompts

Ethics– In AI

Human-Machine Trust

- Moral Principles that govern behavior
- In AI ethics can be influenced (pro and con) by
 - Principles of the developers
 - Principles of the users
 - Evaluation of the technology and Algorithms
 - Policies that govern development and use
 - Availability of AI and computing resources equitably across society
 - Interpretation of Natural Language – issues in reasoning about natural language data
 - AI to supply advice and information vs AI to make decisions
 - Biases or lack of biases in the information supplied
 - Inherent biases, less than complete capabilities, malicious biases



Sources of Positive Ethics and of Ethical Issues– Human-Machine Trust

- Policies
- Development approaches
- Data
- Algorithms
- Users
- Developers
- Capabilities of the AI system
- Inherent or accidental biases
- Unintended consequences
- Automate unethical practices
- Possibility that AI can cause unemployment by taking jobs from humans
- Malicious intent

Ethics of AI - Introduction and
Discussion

- What do you think is the relationship between ethics, trust, and trustworthiness
- What do you think constitutes ethical AI
- What are examples in your world where ethical issues in AI might come up
- How might AI promote ethical decisions – rather than inequitable decisions
- Don't forget to discuss how inequities discuss in human decisions and compare with how AI could be used to amplify those through data and learning or it could be used to mitigate those decision processes



Ethics of AI - Introduction and
Discussion

Some of the Ethics Concerns of Artificial Intelligence Technologies

- How does ethics relate to trust?
- Lack of transparency:
 - Some AI tools or algorithms do not have explanations
 - It may not be clear what data the decision is based on or how the recommendation or decision is made
- Some AI algorithms or decisions may have embedded biases
 - These could be malicious
 - More likely biases may be based on biases that are unrecognized but inherent in the training data (if there is training data)
 - Inherent the way the algorithm makes decisions
 - AI may not have the capability to recognize all of the nuances that human may

Ethics of AI - Introduction and
Discussion

Some of the Ethics Concerns of Artificial Intelligence Technologies

- Concerns for fairness and risk for human rights and gender or racial equity in the decisions
- Data upon which AI decisions are based may not contain equitable balances of ethnic or gender representation
- Equity in the availability of AI technologies – who has access to these advanced technologies



Ethics of AI - Introduction and Discussion

Self-driving Cars

- Imagine all of the near misses that a driver has to deal with
- The AI system may not be able to recognize the difference between a human and an animal
- May not be able to envision the trajectory of a car coming toward you
- Imagine an autonomous car with broken brakes going at full speed toward potential accidents – which way to steer



Ethics of AI - Introduction and Discussion

- Projection Bias – Don't presume that this person thinks like you do
- Base-rate Bias - In different cultures the base rate may be different
- What about Cultural Issues in AI
 - Is the data used to train a machine learning model biased (does it come from people or behaviors of this culture)
 - Knowledge from Experts in AI systems – does it consider the knowledge and behaviors from people of this culture

- The use of AI in judicial systems is increasing
- Can increase the efficiency and accuracy of lawyers in both counselling and litigation
- Existing software systems for judges can be enhanced through AI tools to support judges in drafting new decisions
- How to assure that decisions are, just, accurate or appropriate.
- Is it susceptible to inaccuracies, discriminatory outcomes, embedded or inserted bias and limitations of the learning process?



Ethics of AI - Introduction and Discussion

- What is your opinion
- How will you use information about trust?
- Will you insert it into your crowdsourcing?